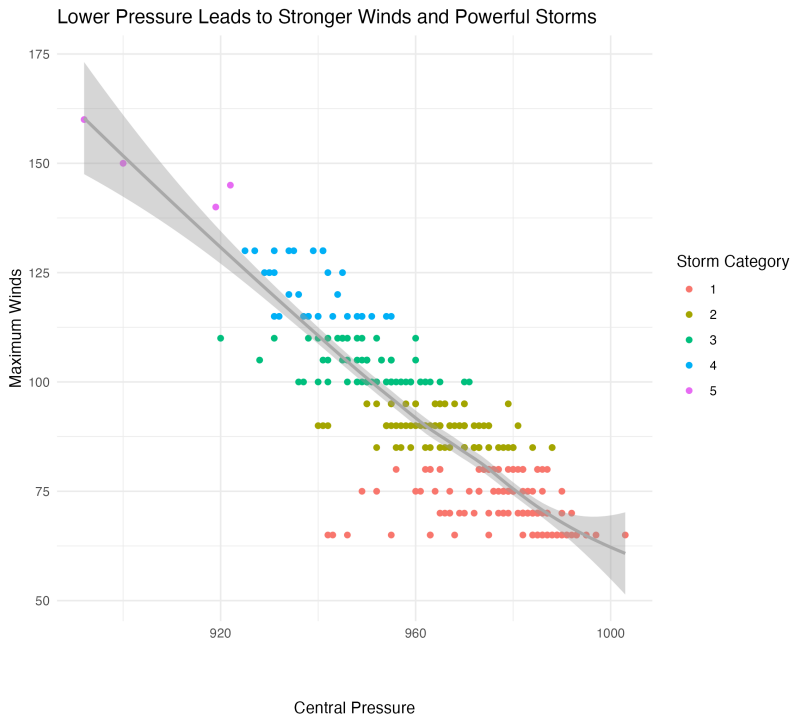
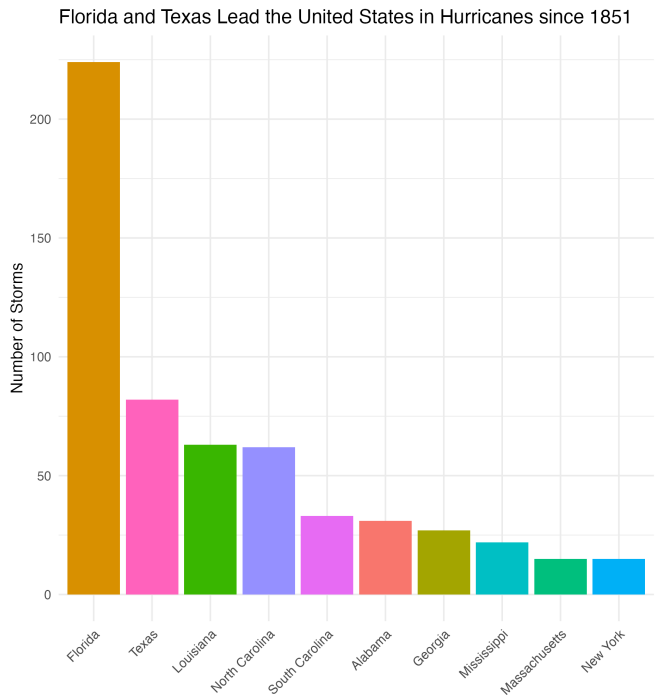
Trixie Southwood 4/27/25





Source: Detailed List of Continental United States Hurricane Impacts/Landfalls. Retrieved from `https://www.aoml.noaa.gov/hrd/hurdat/UShurrs_detailed.html`

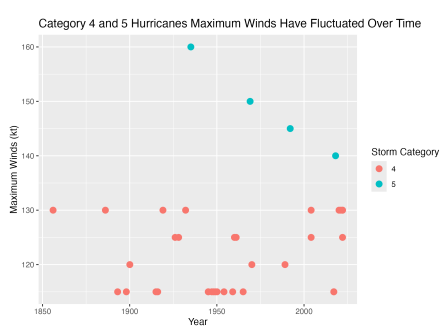
Strong winds from low pressures creates strong hurricanes which largely impact southeastern states

Stronger hurricanes form with lower central pressure which cause stronger winds, these storms are most common in the coastal south east states of America; but there is no correlation between maximum wind over time.

Strong winds generated by low-pressure systems are the key elements in the formation of high category hurricanes, which have a significant impact on the Southeastern states of the U.S. Additionally, twice as many hurricanes made landfall in Florida than any other state in America since 1851.

The data for this analysis comes from the National Oceanic and Atmospheric Administration (NOAA) website: "Detailed List of Continental United States Hurricane Impacts/Landfalls 1851-1970, 1983-2023." Notice the gap from 1970 to 1983, which may indicate a shift in data collection practices or availability during that period.

The scatter plot shows a downward trend between the central pressure and maximum winds. The strongest hurricanes, have low pressure and high winds. The correlation between central pressure and high winds is a causation relationship, as strong wind forms when traveling from high to low pressures. Lower pressure creates a larger change in pressure and therefore stronger winds.



Source: Detailed list of hurricane impacts/landslides in the continental United States.

By isolating category four and five hurricanes, no clear trend in maximum wind speeds was observed over time. This was unexpected as consistent wind intensities would be expected for these high-strength storms, suggesting that other factors may need to be explored in future analyses. The category four and five hurricane graph also highlights the rarity of category five hurricanes; since 1851 there have only been four category five hurricanes in the United States.

The bar chart illustrates that hurricanes are most common along the Southeastern coast of the United States, with Florida having more than twice as many landfalls as any other state. The states with the highest frequency of hurricanes are particularly vulnerable to future storm impacts, showing the critical need for preparation and response strategies in these areas. As hurricane activity continues to evolve, it is important to focus on strengthening resilience in these high risk regions.